PAPUA NEW GUINEA: PUBLIC AND EXTERNAL DEBT SUSTAINABILITY

This annex summarizes the debt sustainability analysis (DSA) for Papua New Guinea using the new framework for low-income countries.[1] The baseline macroeconomic scenario underlying this DSA is the medium-term outlook discussed in the staff report, including the sustained implementation of the ongoing medium-term fiscal strategy. This annex also assesses the impact of various exogenous shocks on the sustainability of public sector debt and external debt under the baseline scenario.

A. Main Assumptions

- Real GDP growth is anticipated to rise gradually from about 3 percent in 2005 to about 4.5 percent by 2010, as agricultural productivity improves, the gas pipeline project and the Ramu nickel mine come on line toward the end of the decade, and structural reform is implemented.

- Monetary policy remains geared toward achieving the objective of keeping inflation at subdued levels and maintaining a sound external position.

- Near balance of the fiscal accounts are expected from 2007 forward, along with gradual improvements in the nonmineral overall and primary balances between 2006 and 2010.

- Papua New Guinea would have access to external concessional finance.

- Prices of key commodities are in line with WEO projections.

- Structural reforms, including those under the Medium-term Development Strategy, would continue to gradually improve governance, law and order, transportation infrastructure, and public sector enterprises' finances. Overall, the investment climate would continue to slowly be enhanced.

B. Total Public Sector Debt Sustainability

The public sector debt ratio has steadily declined from 76 percent of GDP at end-2002 to an estimated 49 percent of GDP at end-2005. The improvement reflects the government's prudent fiscal policy, favorable mineral sector developments, the appreciation of the kina, declining interest rates, and a sustained economic recovery since the current government took office in mid-2002. Multilateral debt accounts for about 58 percent of total external public debt.

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[1] See *Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations* (SM/05/109).

Under the baseline projected scenario, the public sector debt ratio is expected to decline gradually to about 43 percent of GDP by end-2010 and 17 percent by 2025 (Annex VI, Table 1). The reduction would result mainly from continued adherence to the medium-term fiscal strategy and improved growth prospects in line with the assumptions above. The downward debt dynamics are delayed by the assumption that the government borrows about 4 percent of GDP in 2007 to complete the financing of its equity participation in the Highlands-Queensland gas pipeline.[1] In addition, the debt stock from 2005 forward includes contingent liabilities of the government for the first time (debt of statutory authorities). The NPV of public debt would decline from 54 percent of GDP in 2005 to 42 percent of GDP in 2010 and 16 percent of GDP in 2025.The public sector debt service to revenue ratio would remain broadly stable at about 15 percent during the medium term, and falling significantly over the longer term. Public sector debt projections would be affected should contingent liabilities arise in relation to government guarantees (e.g., on public enterprise investment projects) or if the government chose a smaller or higher level of equity investment in the gas pipeline project.

The overall results suggest that, under the IMF-WB DSA framework for LICs, Papua New Guinea's total public sector debt is at moderate-to-high risk of debt distress.[2] The main findings include: (i) only one indicator—NPV of debt to GDP ratio—is above its threshold under the baseline scenario—and it declines under the threshold over the long term, (ii) other indicators are well below their threshold over the entire projected period, and (iii) only under one bound test for lower real GDP growth are these thresholds significantly breached.

The standard sensitivity analysis shows that the path of the debt ratios are particularly sensitive to changes in real GDP growth and the exchange rate (Annex VI, Table 2 and Figure 1). To test the robustness of the baseline scenario, we subjected it to a number of stress tests. Compared to 2005, a decline in real GDP growth to one standard deviation below its historical average would raise the NPV of public debt to GDP ratio by about 16 percentage points of GDP in 2010. A one-time 30 percent real exchange rate depreciation in 2006 would raise the NPV of debt to GDP ratio by about 10 percentage points in 2010.

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[1] A decision has not yet been made on how this borrowing would be financed. As a working assumption, the staff assumes one third would be financed through external concessional loans and the remainder through external commercial borrowing.

[2] Papua New Guinea is rated as a poor performer with regard to its policies and institutions under the joint IMF-WB DSA framework for LICs. Thus, the thresholds applied to Papua New Guinea are: (i) 100 percent for NPV of debt-to-exports, (ii) 30 percent for NPV of debt-to-GDP, and (iii) 200 percent for NPV of debt-to-fiscal revenue; and for debt service: (i) 15 percent of exports and (ii) 25 percent of revenues.

C. External Debt Sustainability

Papua New Guinea's total external debt has also declined steadily in recent years, from 76 percent of GDP in 2002 to about 38 percent of debt in 2005 (Annex VI, Table 3). External debt as a percent of exports has also declined from 19 percent in 2002 to 11 percent in 2005.

External public debt dynamics are sustainable under the baseline scenario, as all three debt indicators (the NPV of public external debt to GDP, the NPV of debt in percent of exports, and the debt service to exports ratio) do not breach the key thresholds. The NPV of public external debt rises in 2007 with the borrowing by about 4 percent of GDP to complete the assumed financing of its equity participation in the Highlands-Queensland gas pipeline. Thereafter it declines, reaching 23 percent of GDP in 2010 and 10 percent of GDP in 2025.

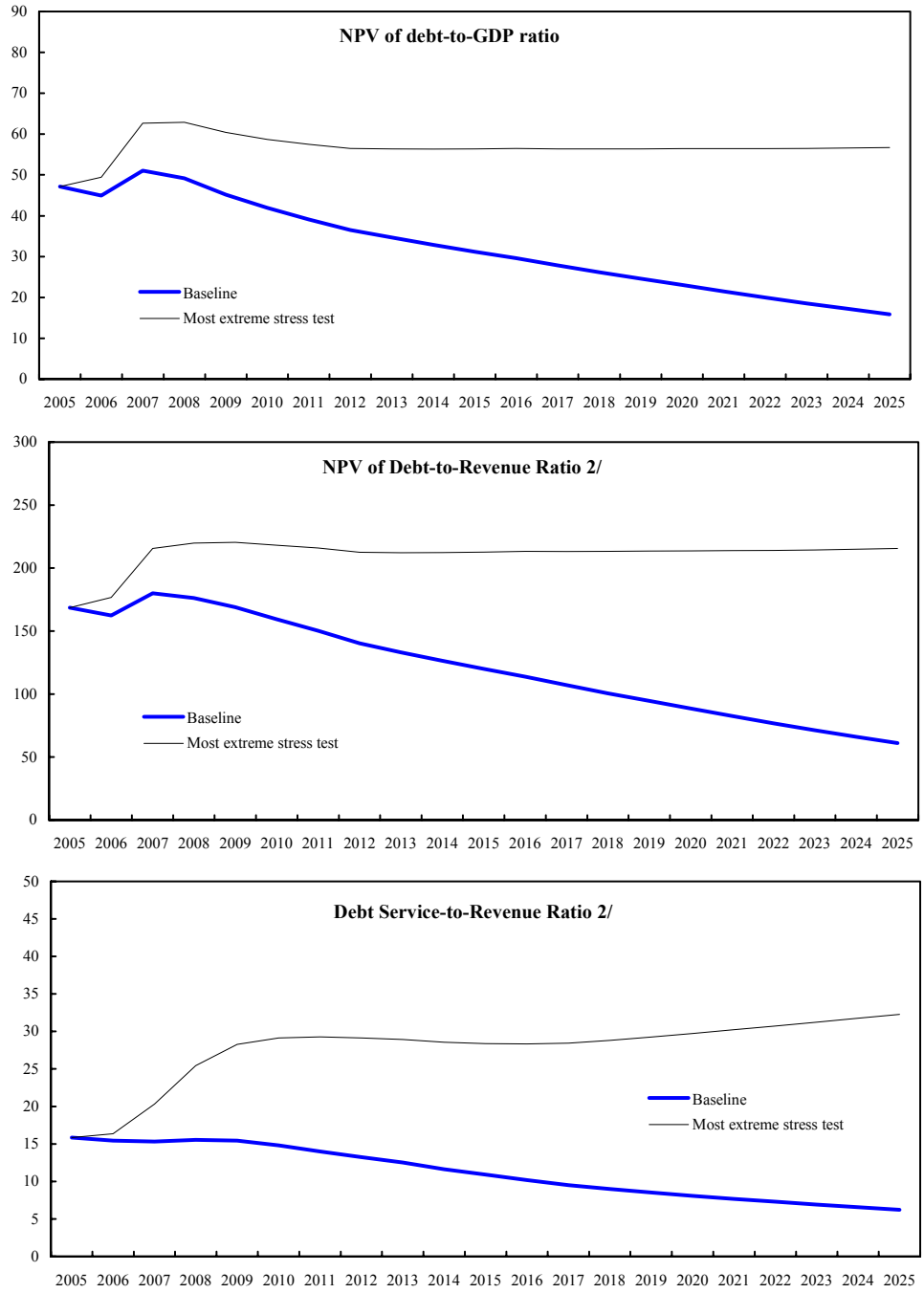
External debt sustainability is most vulnerable to shocks to real GDP growth rates and lower export growth. Stress tests indicate the NPV of public external debt to GDP threshold would initially rise significantly under a lower real GDP growth, lower export growth and exchange rate depreciation shocks, before falling below the thresholds by the end of the period. A decline in real GDP growth by one standard deviation below its historical average would raise the NPV of external public debt from 2005 by 30 percentage points of GDP in 2007 with the impact declining gradually over the medium term to 43 percent of GDP by 2010 and 19 percent by 2025. A decline in export value growth by one standard deviation below its historical average would raise the NPV of external debt-to-GDP ratio from 2005 by 35 percentage points to a high of 61 percent of GDP in 2007, before falling to 19 percent of GDP by 2025 over the longer term. Nonetheless, Papua New Guinea's external debt distress risk is moderated by the fact it has an excellent external repayment record.[3]

D. Conclusions

With continued adherence to prudent macroeconomic and fiscal policies, Papua New Guinea's debt indicators are projected to improve over the medium-to-long term, but slippage of these policies could have adverse consequences on public external debt sustainability. Papua New Guinea's recent progress in reducing the stock of public debt and its capacity to repay future obligations suggest that the stock of debt and debt service as presently projected appear manageable and sustainable. However, this assessment rests on the need to avoid policy slippages and highlights the vulnerability of the debt dynamics to potential shocks. To further help guard against these vulnerabilities, Papua New Guinea needs to ensure that any external borrowing is obtained on concessional terms.

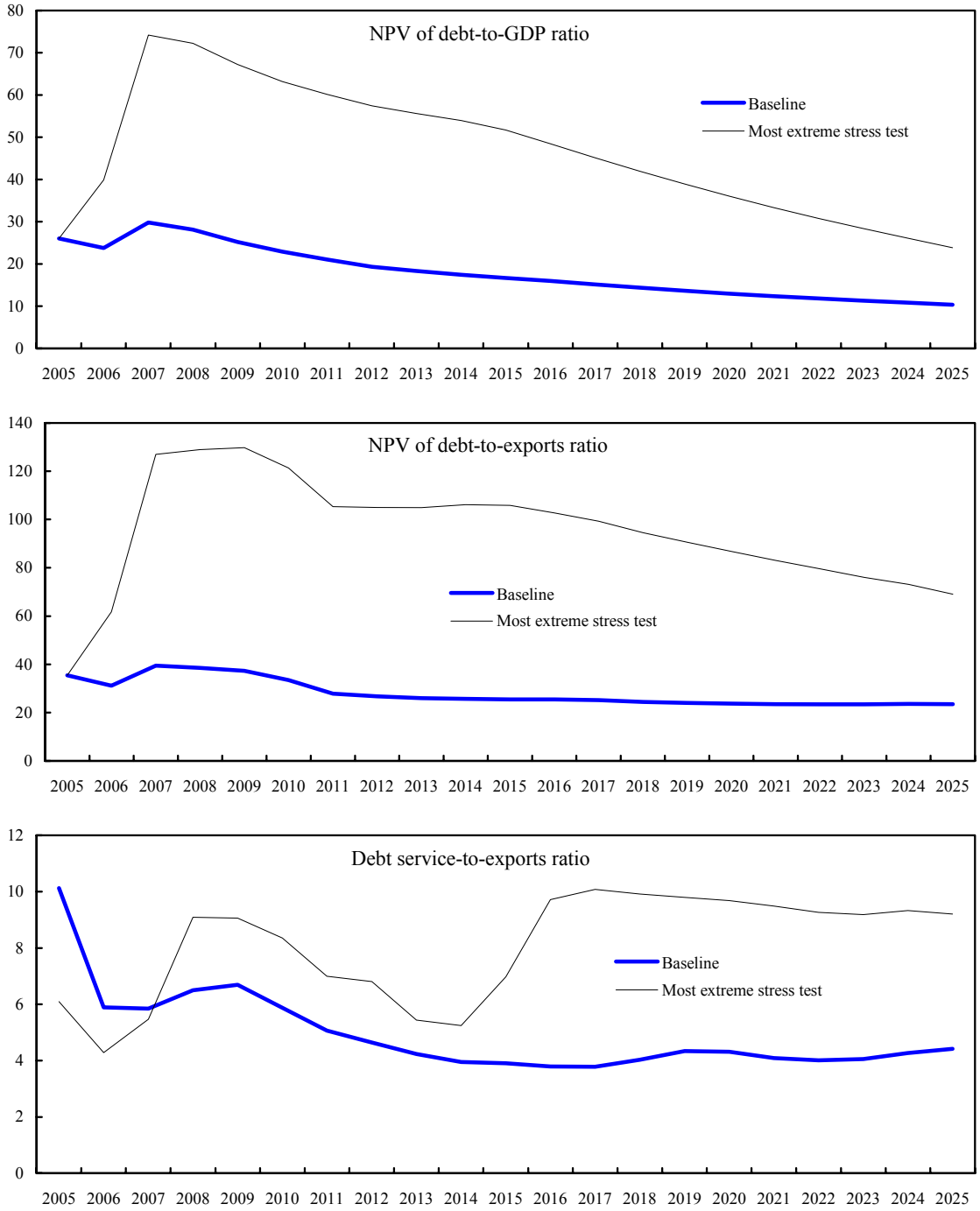
[3] Papua New Guinea has never defaulted on external debt to its creditors.

Figure 1. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2005-2025 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.

Figure 2. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2005-2025
(In percent)



Source: Staff projections and simulations.

Annex VI, Table 1. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002-2025
(In percent of GDP, unless otherwise indicated)

						Estimate	Projections								
	2002	2003	2004	Historical Average 5/	Standard Deviation 5/	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
Public sector debt 1/	76.2	65.0	55.0			48.7	46.3	50.4	49.1	45.7	42.9		31.7	17.4	
o/w foreign-currency denominated	50.2	38.7	32.3			27.4	25.3	29.4	28.3	25.9	24.1		17.3	11.9	
Change in public sector debt	0.4	-11.3	-10.0			-6.3	-2.4	4.1	-1.3	-3.5	-2.7		-2.0	-1.1	
Identified debt-creating flows	-1.4	-13.8	-8.0			-4.9	-0.3	1.0	0.8	-1.7	-1.8		-1.1	-0.6	
Primary deficit	1.4	-3.8	-3.8	-1.7	2.0	-4.8	-2.9	-2.7	-2.5	-2.4	-2.6	-3.0	-1.7	-1.0	-1.5
Revenue and grants	27.8	28.4	31.3			28.0	27.7	28.4	27.9	26.8	26.3		26.0	26.0	
of which : grants	6.0	5.8	6.2			3.2	4.2	4.9	4.8	4.5	4.2		3.3	2.0	
Primary (noninterest) expenditure	29.1	24.6	27.5			23.2	24.8	25.6	25.4	24.3	23.7		24.3	25.1	
Automatic debt dynamics	-1.1	-9.7	-4.0			-3.2	-0.1	1.3	1.1	-1.2	-0.9		-0.2	-0.1	
Contribution from interest rate/growth differential	0.6	0.7	-1.1			-1.6	-0.2	0.5	0.2	-0.9	-0.8		-0.2	-0.1	
of which : contribution from average real interest rate	0.5	2.8	0.8			0.0	1.5	2.1	1.9	1.0	1.1		0.9	0.5	
of which : contribution from real GDP growth	0.1	-2.2	-1.8			-1.6	-1.7	-1.6	-1.7	-1.9	-2.0		-1.1	-0.6	
Contribution from real exchange rate depreciation	-1.7	-10.4	-2.9			-1.5	0.1	0.8	0.9	-0.3	-0.1		...	...	
Other identified debt-creating flows	-1.7	-0.3	-0.2			3.1	2.7	2.5	2.3	2.0	1.7		0.9	0.4	
Privatization receipts (negative)	-1.7	-0.3	-0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			3.1	2.7	2.5	2.3	2.0	1.7		0.9	0.4	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (domestic arrears, court orders, others)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.8	2.6	-2.0			-1.4	-2.1	3.0	-2.1	-1.8	-1.0		-0.9	-0.4	
NPV of public sector debt	75.0	62.5	54.3			47.1	44.9	51.1	49.2	45.2	41.9		31.2	15.9	
o/w foreign-currency denominated	49.0	36.3	31.6			25.9	23.9	30.0	28.3	25.5	23.1		16.8	10.4	
o/w external	49.0	36.3	31.6			25.9	23.9	30.0	28.3	25.5	23.1		16.8	10.4	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	27.2	22.5	17.2			13.6	14.5	14.3	14.0	12.8	11.5		8.4	4.3	
NPV of public sector debt-to-revenue ratio (in percent) 3/	270.1	219.9	173.3			168.5	162.3	180.0	176.1	168.8	159.4		120.0	61.0	
o/w external	176.3	127.6	100.8			92.7	86.3	105.9	101.5	95.1	87.9		64.5	40.0	
Debt service-to-revenue ratio (in percent) 3/ 4/	27.5	34.6	18.1			15.8	15.4	15.3	15.5	15.4	14.8		10.9	6.2	
Primary deficit that stabilizes the debt-to-GDP ratio	1.0	7.4	6.1			1.5	-0.5	-6.8	-1.3	1.1	0.1		0.2	0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-0.2	2.9	2.9	0.6	3.9	3.0	3.5	3.5	3.5	4.1	4.5	3.7	3.5	3.5	3.5
Average nominal interest rate on forex debt (in percent)	3.8	2.9	2.8	4.1	0.9	2.7	2.5	2.7	2.3	2.2	2.4	2.5	2.0	1.0	1.8
Average real interest rate on domestic currency debt (in percent)	-2.2	9.5	2.7	2.0	4.5	-0.4	6.5	9.4	9.1	4.8	5.6	5.8	6.1	10.6	6.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.2	-21.0	-7.7	3.9	13.1	-4.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.3	7.2	4.2	9.0	4.3	6.6	2.8	0.4	0.7	5.0	4.0	3.3	3.5	3.5	3.5
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.1	-13.1	14.9	3.0	13.5	-13.2	10.8	7.1	2.5	-0.2	1.7	1.4	4.0	3.7	3.9
Grant element of new external borrowing (in percent)	...	...	0.0	0.0		0.3	0.4	0.2	0.4	0.4	0.4	0.4	0.3	0.3	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ General government gross debt. The increase in debt in 2007 reflects the staff's understanding regarding current plans for financing a full equity acquisition in the gas pipeline project.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Annex VI, Table 2. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt 2005-2025

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of Debt-to-GDP Ratio								
Baseline	47	45	51	49	45	42	31	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	47	56	56	54	53	49	37
A2. Primary balance is unchanged from 2004	48	45	50	47	42	38	22	-8
A3. Permanently lower GDP growth 1/	47	45	52	51	48	46	42	48
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	47	49	63	63	60	59	56	57
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	47	48	56	54	49	46	35	19
B3. Combination of B1-B2 using one half standard deviation shocks	47	49	60	57	53	49	36	18
B4. One-time 30 percent real depreciation in 2006	47	56	63	61	56	52	40	21
B5. 10 percent of GDP increase in other debt-creating flows in 2006	47	53	58	56	52	48	37	20
NPV of Debt-to-Revenue Ratio 2/								
Baseline	168	162	180	176	169	159	120	61
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	168	170	195	198	198	197	181	136
A2. Primary balance is unchanged from 2004	171	162	177	169	158	146	83	-32
A3. Permanently lower GDP growth 1/	168	164	184	183	179	173	158	183
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	168	177	215	220	220	218	213	216
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	168	172	197	192	184	175	134	72
B3. Combination of B1-B2 using one half standard deviation shocks	168	176	208	202	193	183	137	69
B4. One-time 30 percent real depreciation in 2006	168	203	222	218	209	198	153	82
B5. 10 percent of GDP increase in other debt-creating flows in 2006	168	192	206	201	194	184	142	78
Debt Service-to-Revenue Ratio 2/								
Baseline	16	15	15	16	15	15	11	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	16	18	20	21	21	19	15
A2. Primary balance is unchanged from 2004	16	17	15	13	12	10	2	-11
A3. Permanently lower GDP growth 1/	16	16	16	17	18	18	19	30
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	16	16	20	25	28	29	28	32
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	16	15	21	24	21	18	12	8
B3. Combination of B1-B2 using one half standard deviation shocks	16	16	21	23	20	18	12	7
B4. One-time 30 percent real depreciation in 2006	16	16	17	17	17	17	12	7
B5. 10 percent of GDP increase in other debt-creating flows in 2006	16	15	34	26	21	19	13	9

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Annex VI, Table 3. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2002-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Estimate	Projections									
	2002	2003	2004			2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2020	2025	2011-25 Average
External debt (nominal) 1/	**76.0**	**63.3**	**49.3**			**37.8**	**32.1**	**35.5**	**36.8**	**35.9**	**33.4**	**35.2**	**26.9**	**23.7**	**21.9**	**25.5**
o/w public and publicly guaranteed (PPG)	**51.5**	**44.1**	**34.5**			**27.6**	**25.2**	**29.1**	**28.1**	**25.7**	**23.9**	**26.6**	**17.1**	**13.3**	**11.9**	**15.5**
Change in external debt	2.7	-12.7	-14.0			-11.5	-5.7	3.4	1.3	-0.9	-2.4		-1.5	-0.9	-0.3	-0.8
Identified net debt-creating flows	2.4	-20.1	-12.5			-8.5	-7.6	-1.2	-2.2	-3.2	-6.9		-7.7	-2.4	3.5	-4.6
Non-interest current account deficit	**-2.3**	**-6.1**	**-3.4**	**-6.8**	**6.3**	**-5.3**	**-5.7**	**0.9**	**1.6**	**0.4**	**-4.8**	**-2.2**	**-6.7**	**-1.5**	**4.1**	**-3.6**
Deficit in balance of goods and services	14.5	7.5	8.9			7.8	6.3	14.5	15.3	13.6	8.3		5.8	10.2	14.8	8.4
Exports	60.3	65.9	64.5			73.6	76.4	75.6	72.8	67.7	68.4		65.3	54.6	44.1	59.1
Imports	74.7	73.4	73.4			81.4	82.7	90.1	88.1	81.3	76.8		71.1	64.8	58.9	67.5
Net current transfers (negative = inflow)	-6.5	-6.5	-4.2	-4.8	1.3	-4.1	-4.7	-5.4	-5.2	-4.8	-4.5		-3.4	-2.6	-2.0	-3.0
Other current account flows (negative = net inflow)	-10.2	-7.1	-8.1			-9.1	-7.3	-8.3	-8.5	-8.4	-8.6		-9.1	-9.1	-8.7	-9.0
Net FDI (negative = inflow)	**-0.6**	**-2.7**	**-0.6**	**-3.7**	**3.4**	**-1.4**	**-1.7**	**-2.6**	**-4.2**	**-3.7**	**-2.1**	**-2.6**	**-1.1**	**-0.9**	**-0.7**	**-1.1**
Endogenous debt dynamics 2/	**5.3**	**-11.3**	**-8.4**			**-1.8**	**-0.2**	**0.5**	**0.4**	**0.2**	**0.0**	**-0.2**	**0.1**	**0.0**	**0.0**	**0.1**
Contribution from nominal interest rate	3.3	1.7	1.4			1.2	1.1	1.2	1.1	0.9	0.9		0.6	0.5	0.5	0.6
Contribution from real GDP growth	10.0	-7.9	-7.2			-2.9	-1.3	-0.7	-0.6	-0.8	-0.9		-0.5	-0.5	-0.4	-0.5
Contribution from price and exchange rate changes	-8.0	-5.1	-2.6			…	…	…	…	…	…		…	…	…	
Residual (3-4) 3/	**0.3**	**7.4**	**-1.5**			**-3.0**	**1.9**	**4.6**	**3.5**	**2.3**	**4.5**	**2.3**	**6.2**	**1.5**	**-3.8**	**3.9**
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0
NPV of external debt 4/	**69.9**	**56.8**	**47.4**			**36.3**	**30.7**	**36.2**	**36.8**	**35.4**	**32.4**	**34.6**	**26.4**	**23.4**	**20.4**	**24.7**
In percent of exports	115.9	86.3	73.6			49.3	40.3	47.8	50.5	52.2	47.4		40.4	42.8	46.2	42.2
NPV of PPG external debt	**45.4**	**37.6**	**32.5**			**26.1**	**23.8**	**29.8**	**28.1**	**25.2**	**22.9**	**26.0**	**16.7**	**13.0**	**10.4**	**14.8**
In percent of exports	75.3	57.1	50.5			35.4	31.1	39.5	38.6	37.2	33.5		25.5	23.7	23.5	24.8
Debt service-to-exports ratio (in percent)	19.3	22.6	15.1			10.8	6.6	6.7	7.2	7.4	6.5		4.3	4.7	5.2	4.7
PPG debt service-to-exports ratio (in percent)	**16.1**	**21.5**	**14.3**			**10.1**	**5.9**	**5.8**	**6.5**	**6.7**	**5.9**	**6.8**	**3.9**	**4.3**	**4.4**	**4.2**
Total gross financing need (billions of U.S. dollars)	0.3	0.2	0.2			0.1	-0.1	0.2	0.1	0.1	-0.2		-0.4	0.0	0.8	-0.1
Non-interest current account deficit that stabilizes debt ratio	-5.0	6.5	10.6			6.2	0.0	-2.5	0.3	1.3	-2.3		-5.2	-0.6	4.4	-2.8
Key macroeconomic assumptions																
Real GDP growth (in percent)	-0.2	2.9	2.9	0.6	3.9	3.0	3.5	3.5	3.5	4.1	4.5	3.7	3.5	3.5	3.5	3.1
GDP deflator in US dollar terms (change in percent)	12.3	7.2	4.2	9.0	4.3	6.6	2.8	0.4	0.7	5.0	4.0	3.3	3.5	3.5	3.5	3.5
Effective interest rate (percent) 5/	4.4	2.7	2.5	3.7	0.8	2.7	3.1	3.7	3.0	2.8	2.7	3.0	2.2	2.2	2.2	2.3
Growth of exports of G&S (US dollar terms, in percent)	-16.5	31.8	15.8	0.5	15.9	30.0	10.5	1.6	-1.1	0.0	7.9	8.1	1.5	1.6	1.4	2.5
Growth of imports of G&S (US dollar terms, in percent)	-0.4	18.5	18.3	3.3	15.9	26.2	8.2	11.9	0.3	-0.8	0.8	7.8	3.6	3.6	3.5	3.7
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	34.7	39.5	20.5	39.2	38.1	39.4	35.2	32.4	32.0	31.5	32.4
Memorandym item:																
Nominal GDP (billions of US dollars)	3.0	3.6	4.3			4.9	5.2	5.3	5.5	5.9	6.3		8.2	10.7	14.0	

Source: Staff simulations.

1/ Includes both public and private sector external debt. The increase in debt in 2007 reflects the staff's understanding regarding current plans for financing a full equity acquisition in the gas pipeline project.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Annex VI, Table 4. Papua New Guinea: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of debt-to-GDP ratio								
Baseline	26	24	30	28	25	23	17	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	26	22	22	14	7	3	0	0
A2. New public sector loans on less favorable terms in 2006-25 2/	26	24	33	32	29	27	22	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	26	33	56	53	47	43	31	19
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	26	36	61	60	56	53	44	19
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	26	23	28	26	24	22	16	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	26	26	35	33	30	27	21	12
B5. Combination of B1-B4 using one-half standard deviation shocks	26	40	74	72	67	63	52	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	26	34	42	40	36	32	24	15
NPV of debt-to-exports ratio								
Baseline	35	31	39	39	37	33	26	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	35	28	29	20	10	4	0	0
A2. New public sector loans on less favorable terms in 2006-25 2/	35	31	43	43	43	40	33	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	35	31	39	39	37	33	26	24
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	35	62	127	129	130	121	106	69
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	35	31	39	39	37	33	26	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	35	34	46	45	44	40	32	27
B5. Combination of B1-B4 using one-half standard deviation shocks	35	52	92	93	93	87	75	51
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	35	31	39	39	37	33	26	24
Debt service ratio								
Baseline	10	6	6	6	7	6	4	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	6	3	3	4	4	3	1	0
A2. New public sector loans on less favorable terms in 2006-25 2/	6	3	3	4	4	4	4	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	6	3	3	5	5	4	2	2
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	6	4	5	9	9	8	7	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	6	3	3	5	5	4	2	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	6	3	3	5	5	4	3	3
B5. Combination of B1-B4 using one-half standard deviation shocks	6	4	4	7	7	7	5	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	6	3	3	5	5	4	2	2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	31	31	31	31	31	31	31	31

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt (FDI)creating flows. The NPV of debt and de service tend to zero over time as the 10-year average of the current account balance (a 6.8 percent of GDP surplus) and net FDI inflows imply that debt is paid off.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.